

Alyssa Johnson

Quantitative Developer | Physics PhD candidate building ML systems for signal detection, forecasting, and alpha-driven financial modeling.
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SKILLS | Cloud, Infrastructure & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, DevOps, Cloud Run
Programming & Systems: Python (Pandas, NumPy, scikit-learn, PyTorch), SQL, JavaScript, TypeScript, Linux, Git, Docker, APIs, Distributed/Parallel Computing, OpenShift, Flask, MATLAB, React, HTML/CSS, Grafana, Guacamole, *some* Fortran
ML / AI: Time Series Forecasting, Statistical Modeling, Tree-Based & Ensemble Models (XGBoost, LightGBM), Deep Learning, NLP/LLMs, TensorFlow, RAG Systems, Vector Embeddings, Pinecone, Hugging Face, Agentic Systems
Quantitative Finance: Derivatives (Options, Futures, Swaps), Equities, Fixed Income, Commodities, Cryptocurrencies, Bloomberg Terminal / OpenBB, Bayesian Modeling, A/B Testing Methodologies, Anomaly Detection

EXPERIENCE

UCSD Triton Quantitative Trading Club - Lead, Strategy Development Research Group March 2026 - present

- **Project**: Building QPulse, a cloud-native agentic platform ingesting financial news, SEC filings, and market data to generate multi-asset signals (equities, fixed income, crypto, derivatives) via LLM-powered RAG pipelines; deployed on AWS/GCP with Docker/Kubernetes, served through a Flask API and React/TypeScript dashboard.
- **Project**: Constructing a suite of delta-hedged options strategies in QuantConnect, implementing vega-scaled positioning, IV percentile filtering, and automated Greeks-based entry and exit logic across long/short vega and gamma regimes on SPY. 🌐
- **Project**: Constructed an in-database ML forecasting pipeline using MySQL HeatWave, benchmarking time series model performance against external Python baselines; documenting tradeoffs in latency, accuracy, and scalability. 🌐

UCSD Optical & Infrared Laboratory - Graduate Student Researcher April 2025 - present

- **Award** → High Energy Physics Consortium for Advanced Training Fellowship → *fellowship is ongoing*
- **Award** → University of California, San Diego Sloan Scholars Fellowship → *fellowship is ongoing*
- **Project**: (PhD Thesis) Assembled and validated telescope modules and SiPM detector systems for nanosecond-scale optical transient detection and high-time-resolution instrumentation. 🌐
- **Project**: Developed a Python full-sky figure-of-merit pipeline (3M+ directions) combining stellar density and dust ext. 🌐

UCSD Cosmology Group - Graduate Student Researcher August 2022 - April 2025

- **Award** → GEM Fellowship Program
- **Project**: (M.S. Thesis) Developed statistical signal processing and noise-modeling pipelines for CMB detector time-series data, contributing to the Simons Observatory instrumentation stack (co-authored SPIE 2024). 🌐

NASA Goddard Space Flight Center - NASA GSFC Intern (x3) Aug. 2021 - Aug. 2022

- **Awards** → NASA Sally Ride Scholarship, NASA Space Generation Advisory Council Scholarship
- **Main Project**: Delivered single-user relay and handover experiments that demonstrated user data services for optical communications, defining spacecraft configurations and scheduling parameters to support LCRD mission operations.

Cal Poly, Humboldt, Gravitational Physics Lab - Student Researcher February 2019 - July 2022

- **Award** → Cal-Bridge Program Scholarship
- **Project**: Optimized an active servo-leveling control system for a precision short-range gravity experiment using numerical modeling and sensor feedback analysis; published in J. Undergrad. Rep. Phys. (2023). 🌐

Montana State University Solar Physics REU - Student Researcher 🌐 June - August 2021

EDUCATION

- University of California, San Diego | **PhD Candidate in Astrophysics** Expected Spring 2028
 - University of California, San Diego | **MSc in Astrophysics** grad. June 2025
 - California State Polytechnic University, Humboldt | **BSc in Physics, Cum Laude** grad. Dec. 2021
 - Additional degrees: The Ohio State University, MFA (grad. 2014); Temple University, BFA (grad. 2011)
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SELECT PUBLICATIONS

- **Johnson, A.**, Rogers, C., Dunkley, N., Gengo, M., Hoyle C.D., *Optimization of an Active Leveling Scheme for a Short-Range Gravity Experiment.* J. Undergrad. Rep. Phys. **33**, 100002 (2023)
- Bhimani, Sanah, et al. (including **Johnson, A.**). "The Simons Observatory: Deployment and current configuration of the Observatory Control System for SAT-MF1 and data access software systems." Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII. Vol. 13102. SPIE, 2024.